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Activity 1: AI-Assisted Code Review

Purpose

Learn how to use AI to perform a structured code review by evaluating code quality, readability, maintainability, coding standards, and potential improvements before code is merged into the project.

Pre-requisites

- Completion of Day 3 activities
- Working SmartBank EMI Calculator application
- VS Code with Cline configured

Scenario

The SmartBank development team has completed the EMI Calculator feature. Before the feature can be merged into the main development branch, every code change must undergo a peer review.

As a developer, your responsibility is to use AI as a code review assistant to identify potential issues, suggest improvements, and verify that the code follows enterprise development standards.

Note: AI should be treated as a review assistant. Always evaluate its suggestions before accepting or implementing any changes.

Tasks

Task 1: Perform an Overall Code Review

Review the complete EMI Calculator application and identify areas that require improvement.

Suggested Prompt

Act as a Senior Software Engineer reviewing this Oracle JET and KnockoutJS application.

Perform a comprehensive code review.

Evaluate the following:

- Overall code quality
- Readability
- Maintainability
- Naming conventions
- Code organization
- Reusability
- Consistency
- Oracle JET and KnockoutJS best practices

Present your findings under:

- Strengths
- Areas for Improvement
- Recommendations

Do not modify the code.

Task 2: Review Business Logic

Analyze the EMI calculation logic to ensure it is correct and reliable.

Suggested Prompt

Review the EMI calculation logic implemented in this application.

Verify:

- Formula correctness
- Variable naming
- Input validation
- Boundary conditions
- Error handling
- Potential edge cases

Identify any logical issues or business risks.

Explain your findings before suggesting improvements.

Do not rewrite the implementation.

Task 3: Review Code Quality and Best Practices

Evaluate whether the code follows common development standards.

Suggested Prompt

Review this Oracle JET application against common software engineering best practices.

Evaluate:

- Duplicate code
- Function size
- Modularity
- Separation of concerns
- Maintainability
- Readability
- Coding standards

Suggest practical improvements without changing the application's functionality.

Task 4: Review User Experience and Validation

Review the application from an end-user perspective.

Suggested Prompt

Review the EMI Calculator from a user's perspective.

Evaluate:

- User input validation
- Error messages
- Labels
- Button names
- Overall usability
- Accessibility considerations
- User experience improvements

Provide recommendations without modifying the UI.

Expected Deliverables

By the end of this activity, you should have:

- Code Review Report
- Business Logic Review
- Best Practice Recommendations
- User Experience Review
- List of suggested improvements